

A Doughnut-Economics Constraint Model of Floor and Ceiling Trade-offs

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The Starting Point: Raworth's Critique of GDP

- The current economic model treats **GDP growth** as the goal every country pursues within the global economic framework.
- Kate Raworth's critique is that the economy is **not a self-contained system**: it is embedded in society, politics, and ecology, and continuously affects all three. A model that measures success only by GDP cannot notice when it is damaging these other dimensions.
- In response, Raworth proposes the **doughnut model**: a country succeeds when it simultaneously
 - guarantees society a set of fundamental benefits, *and*
 - keeps its ecological consumption below the point of no return.

The Two Philosophical Frameworks

The study combines Frankfurt's sufficientarianism, together with Sen and Nussbaum's capability approach, to define the social floor, and Rockström and Steffen's planetary boundaries to define the ecological ceiling.

- **Frankfurt (sufficientarianism):** the morally relevant goal is **sufficiency**, each person having enough for their own needs, rather than equality of resources; chasing equality instead alienates people into status competition.
- **Sen (1999):** development should be judged by **ends** (people's real freedom to live the life they choose) rather than by **means** (GDP, income).
- **Nussbaum (2011):** building directly on Sen, defines **ten central capabilities** owed to every citizen as non-derogable rights. **Consequence for the model:** the direct basis for encoding the social floor as **11 independent hard constraints**, not one aggregate index.
- **Rockström & Steffen:** define nine **planetary boundaries**, non-linear Earth-system processes with tipping points; three (four in the Steffen et al. 2015 update) already crossed. **Consequence for the model:** coupled but non-substitutable limits, hence encoded as **independent, non-negotiable ecological constraints**.

Frankfurt and the Sen-Nussbaum line are two independent philosophical traditions, born of different questions. They converge on the same structural conclusion: a sharp **sufficiency threshold** exists, below which a deficit cannot be compensated by a surplus elsewhere.

Empirical Quantification: O'Neill et al. (2018)

- The doughnut model becomes empirically quantifiable in **2018**, when O'Neill, Fanning, Lamb, and Steinberger collect data from **151 nations** across **7 ecological** and **11 social** dimensions for each country.
- The authors argue that natural resources constitute the primary **physical means** and human well-being the ultimate **end**; the two extremes of the spectrum interact through each state's own technological, social, and infrastructural provisioning systems.
- **No country in the world** lies within the doughnut. Achievement of social goals is linked to resource consumption through a **non-linear saturation relationship**, so that beyond a certain threshold each additional unit of resources consumed produces diminishing marginal social benefit. Satisfying **physical** needs (nutrition, sanitation) requires a markedly smaller biophysical footprint than satisfying **qualitative or perceived** goals (income, life satisfaction).

What this supplies to the ASP model

The 151 countries, the 18 dimensions, and the encoding of every value as a **ratio to a threshold** (≥ 100 for the floor, ≤ 100 for the ceiling) all come directly from this study, which supplies both the empirical dataset and the logical schema of the entire codebase.

- The IPCC Working Group III report synthesizes the scientific consensus on the feasibility of meeting the Paris Agreement targets, outlining decarbonization pathways to stay **within 1.5°C** (2°C is the less ambitious reference limit, not the central target).
- The report also notes the collapse in the cost of low-emission technologies, now competitive with fossil fuels.
- Rapid, deep emissions cuts are needed across all sectors, plus a restructuring of the **AFOLU** sector (agriculture, forestry, land use).
- Global capital to close investment gaps is sufficient, but current flows remain **three to six times** below required levels, penalizing developing countries most.

IEA Net Zero by 2050

- The pathway requires halting, from 2021, approval of **new** fossil-fuel fields and **new** coal plants without emissions-capture systems.
- By 2050, nearly **90%** of global *electricity generation* must come from renewables (70% from solar and wind): this figure refers to the **electricity mix**, not total energy demand, since electricity covers only about half of total final consumption in the NZE scenario.
- By 2030: full universal access to electricity (~**785 million** people) and to clean cooking systems (~**2.6 billion** people).

What these two sources validate in the model

IPCC AR6 WGIII and IEA NZE2050 empirically validate the direction of lever `l1_renewable_energy`, and the explicit conflict the code imposes between it and `l11_fossil_energy_expansion`.

What Is a Lever, and What Its Source Does (Not) Prove

- A lever is a **binary intervention**: selected or not, with no in-between values. If selected, it adds a **fixed delta** to every dimension it touches; the solver only decides *whether* to apply it, never *how much*. Some levers touch a single dimension (e.g. `l13_sanitation_infrastructure`), others more than one (e.g. `l1_renewable_energy` on both CO2 and energy access), and two (`l1` and `l11_fossil_energy_expansion`) are **mutually exclusive**.
- Every lever is accompanied by an “indicative” source, but **none of these sources provides a causal, per-country estimate** of that policy’s effect: no dataset of empirically estimated causal effects exists for these policies. Each source justifies only the **direction** and a **plausible order of magnitude**: the delta itself remains an illustrative assumption, not a measured parameter.
- This is why the project includes `representativeness_check.py`, which makes the assumption **verifiable** instead of leaving it as a stated-but-uncontrolled limitation.

What representativeness_check.py Does

- The script does *not* estimate a lever's effect: it only checks that it is **not implausible**, by comparing it to how much the same dimension already varies naturally among the 151 real countries in the O'Neill et al. dataset.
- For each indicator, it compares how much “normal” countries (25th–75th percentile) differ from each other, and how much countries at the non-anomalous extremes (10th–90th percentile) differ, then asks: *is the jump this lever promises smaller than the jump that already exists between real countries on this indicator?*
- If yes, the lever is “**representative**”: it does not ask a country to make a leap bigger than reality itself shows to already be possible.
- **Result** (report/representativeness_table.md): no lever exceeds the real P10–P90 range. Most are **conservative** (11's effect on CO2 uses only 7% of the true IQR); a smaller group is more **ambitious**, between 45% and 89% of the IQR (16, 19, 18, 112, 17, 15, 114). This does not make the assumptions true, but anchors them to observed variation instead of arbitrary numbers.

The 14 Levers: Ecological / Resource Group

These five levers act primarily on the **ecological ceiling**; 111 is the deliberate “dirty” alternative to 11, and the two are coded as mutually exclusive.

Lever	Effect (Δ)	Source
11_renewable_energy	co2 -35 ; energy_access $+15$	IPCC AR6 WGIII & IEA NZE2050: both validate renewables as high-potential for emissions cuts with simultaneous energy-access expansion.
12_circular_economy	material_footprint -30 ; eco_footprint -15	Ellen MacArthur Foundation (2013): reuse, recycling, extended product life reduce virgin material demand.
13_sustainable_agriculture	nitrogen -25 ; phosphorus -25 ; ehanpp -30 ; nutrition -10 (explicit trade-off)	FAO / EAT–Lancet Commission (Willett et al., 2019): N/P load and land-pressure cuts from less-intensive agriculture, with possible short-term nutritional trade-offs.
14_water_efficiency	blue_water -20	UN-Water (2021), <i>Valuing Water</i> : efficiency infrastructure and demand management on freshwater consumption.
111_fossil_energy_expansion (conflicts with 11)	co2 $+15$; energy_access $+20$	Deliberately modeled as the cheaper but “dirty” alternative to 11, forcing an explicit normative fork.

The 14 Levers: Socio-Economic Floor Group

These five levers act primarily on the **social floor**, spanning income support, health, taxation, and education.

Lever	Effect (Δ)	Source
15_working_time_reduction	eco_footprint -20 ; material_footprint -15 ; life_satisfaction $+22$; employment $+10$	Kallis et al. (2018) on degrowth; Icelandic short-week trial (Autonomy/Alda, 2021): reduced ecological throughput with stable-or-better employment and life satisfaction.
16_universal_basic_income	income $+20$; sanitation $+15$; nutrition $+10$	ILO Recommendation 202/2012 on Social Protection Floors: links minimum-income guarantees to access to essential goods.
17_universal_healthcare	healthy_life_exp $+20$; social_support $+10$	WHO & World Bank (2017), <i>Tracking Universal Health Coverage</i> .
19_wealth_tax	equality $+25$; income $+10$	Piketty, <i>Capital in the Twenty-First Century</i> (2014): the most theoretical/macro source in the table, most exposed to objections in discussion.
110_education_investment	education $+20$; employment $+10$	UNESCO (2015), Incheon Declaration / SDG4 framework: links education investment to downstream employment outcomes.

The 14 Levers: Governance & Special-Case Group

These four levers cover **governance quality** and two special cases: 114 closes both floor and ceiling gaps at once, while 113 is the most direct mapping in the whole table.

Lever	Effect (Δ)	Source
18_participatory_governance	democratic_quality +30; social_support +10	OECD (2020), <i>Catching the Deliberative Wave</i> : effect of participatory budgets and deliberative bodies on perceived democratic legitimacy.
112_decentralization_reform	democratic_quality +20; social_support +5	Ostrom, <i>Governing the Commons</i> (1990): polycentric governance and its effects on collective action and local legitimacy.
113_sanitation_infrastructure	sanitation +25	WHO/UNICEF JMP WASH report (2021): the source literally measures the indicator the lever changes.
114_carbon_fee_dividend_reforestation	co2 -50; ehanpp -35; equality +15	Hansen et al. (2013, <i>PLOS ONE</i>) for fee-and-dividend (equal per-capita dividend explains the equality effect); IPCC AR6 WGIII for reforestation as a carbon sink.

A Six-Step Pipeline

- ❶ `data_prep.py`: transforms the O'Neill et al. data sheet into ASP facts (bio_value/soc_value per country, dimension, value).
- ❷ `levers.lp`: declares the 14 levers and their effects (affects_bio/affects_soc), with the indicative sources discussed in Section 5.
- ❸ `doughnut.lp`: the central program, choosing a subset of levers and checking whether it satisfies floor and ceiling together.
- ❹ `best_effort.lp`: kicks in only when no subset works, measuring how much stays unsatisfied in the best possible combination.
- ❺ `run_scenario.py`: orchestrates everything by calling `clingo` and writes the results used in Section 7.
- ❻ `representativeness_check.py`: separately verifies lever plausibility (already discussed in Section 5).

Where the argument actually lives

Steps 1, 2, 5, 6 are necessary but “service” steps. The real argumentative content, where the report’s philosophical theses become verifiable code, is entirely in `doughnut.lp` and `best_effort.lp`, the focus of the rest of this section.

- Every country is a series of facts `bio_value(country, dimension, value)` and `soc_value(country, dimension, value)`, with values scaled $\times 100$ to work with integers (clingo has no native decimal numbers).
- Every lever is a fact `lever(L)` accompanied by `affects_bio(L, dimension, delta)` and/or `affects_soc(L, dimension, delta)`: a fixed, signed delta, on the same $\times 100$ scale.

doughnut.lp: Choice and Hard Constraints

```
{ select(L) } :- lever(L).

:- select(L1), select(L2), conflicts(L1, L2).

bio_effect(Dim, Sum) :- bio_dim(Dim),
    Sum = #sum { Delta,L : select(L), affects_bio(L, Dim, Delta) }.
soc_effect(Dim, Sum) :- soc_dim(Dim),
    Sum = #sum { Delta,L : select(L), affects_soc(L, Dim, Delta) }.

new_bio_value(C, Dim, V + E) :- target(C), bio_value(C, Dim, V), bio_effect(Dim, E).
new_soc_value(C, Dim, V + E) :- target(C), soc_value(C, Dim, V), soc_effect(Dim, E).

ceiling_ok(C, Dim) :- new_bio_value(C, Dim, V), V <= 100.
floor_met(C, Dim) :- new_soc_value(C, Dim, V), V >= 100.

:- target(C), bio_value(C, Dim, _), not ceiling_ok(C, Dim).
:- target(C), soc_value(C, Dim, _), not floor_met(C, Dim).

#minimize { 1,L : select(L) }.
```

Reading the Mechanics, and Where Philosophy Becomes Code

- **Line 1** (curly braces) is a **choice**: for each of the 14 levers, the solver is free to activate it or not (2^{14} candidate combinations); **line 3** uses `conflicts/2` to discard any combination activating 11 and 111 together. The two `#sum` aggregates sum the deltas of all selected levers, dimension by dimension, and add the result to the country's starting value (`target(C)`), giving the new post-reform value.
- The last two rules before `#minimize` are genuine **headless rules** (`:- body.`), each forbidding its body from being true: one discards any answer where an ecological dimension exceeds 100, the other any answer where a social dimension stays below 100, checked **one dimension at a time** rather than as an aggregate score. This is where **sufficientarianism and strong sustainability become code**: a solution fixing 17 of 18 dimensions is discarded anyway, since no surplus elsewhere can compensate for a single deficit or overshoot.
- Finally, `#minimize` makes the solver prefer, among surviving solutions, those with **fewer active levers**, a parsimony criterion stated explicitly in the code.

best_effort.lp: When No Solution Exists

If no lever combination satisfies both constraints, `doughnut.lp` correctly answers UNSAT, but a bare “no” is not useful for a political or ethical discussion. `best_effort.lp` reuses the exact same facts and effect-summing machinery, but **removes** the two hard constraints and instead minimizes the remaining overshoot/deficit:

```
overshoot(Dim, V - 100) :- new_bio_value(_, Dim, V), V > 100.  
deficit(Dim, 100 - V) :- new_soc_value(_, Dim, V), V < 100.  
  
#minimize { P,Dim : overshoot(Dim, P) }.  
#minimize { P,Dim : deficit(Dim, P) }.
```

- Selecting no lever at all is always a valid (if poor) answer, so this diagnostic model is **always solvable**.
- It is invoked only *after* `doughnut.lp` has already returned UNSAT.

Why These Three Countries

- Among the 151 countries in the O'Neill dataset, **Vietnam, Costa Rica, and the United States** were selected by ranking all countries on the severity of ecological overshoot and social deficit.
- This spans a **spectrum of difficulty**: an easy case, an intermediate case, and a structurally hard case.

Vietnam: Solvable, with Equivalent Alternatives

Baseline: a single ecological overshoot (CO2 at 135) and four social deficits (democratic quality, sanitation, equality, life satisfaction); 17 of 18 dimensions tracked (Vietnam lacks the education datapoint). The problem is **solvable**: `doughnut.lp` finds two optimal models, both six levers, differing only in how CO2 is closed.

Model	Levers	Outcome
1	11, 15, 18, 19, 112, 113	All 17 dims. inside the doughnut
2	15, 18, 19, 112, 113, 114	All 17 dims. inside the doughnut

Both models share a 5-lever core and differ only in `11_renewable_energy` vs. `114_carbon_fee_dividend_reforestation` to close CO2: L14 also cuts eHANPP deeper and lifts equality more (per-capita dividend), but leaves `energy_access` unchanged, unlike L1. The two models are formally equivalent under the doughnut constraints, though their downstream consequences differ.

Costa Rica: Infeasible at 13 Levers, Feasible at 14

Baseline: 6 of 7 ecological dimensions overshoot, with only **two small** social deficits (equality, employment). With the initial 13-lever menu the problem is **infeasible** (best-effort residual violation = 77 points, still failing CO2/eHANPP/equality). Adding the 14th lever (114, fee-and-dividend + reforestation) makes it **solvable**, with **four** equivalent optimal models of **eight levers** each, sharing a six-lever core:

Model	Core (11, 12, 13, 15, 19, 114) + two of
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1	16 (UBI), 112 (decentralization)
---	----------------------------------

2	16 (UBI), 18 (participatory governance)
---	---

3	18 (participatory governance), 113 (sanitation)
---	---

4	112 (decentralization), 113 (sanitation)
---	--

The four solutions are **formally equivalent** w.r.t. the doughnut constraints but build **different societies**: participatory governance vs. top-down decentralization reflect different ideas of democratic legitimacy. The model correctly leaves this choice to human political judgment instead of resolving it computationally.

United States: Structurally Infeasible

Baseline: overshoots **all seven** ecological dimensions, some by an order of magnitude (**CO2 at 1314**, over 13× the sustainable share), while the social floor is already **almost entirely met**.

- Even with all **14** levers available, the problem remains **infeasible**: there is no optimal, constraint-satisfying solution to tabulate, only a best-effort diagnostic.
- The best-effort model selects **8 levers** (11, 12, 13, 14, 15, 19, 110, 114), the largest and most varied set of the three case studies. This fully closes the social floor (11/11 dimensions met) but leaves **5 of 7** ecological ceilings breached, for a **residual violation of 2815 points** (~ 36× Costa Rica's pre-L14 figure). The impossibility concentrates in CO2, phosphorus, and nitrogen, where US overshoot is most extreme to begin with.

What this reproduces

This reproduces, as a logical consequence of the model rather than a restatement of the paper's results, O'Neill et al.'s (2018) central finding: no country satisfies the social floor without breaching the ecological ceiling.

What the ASP Formulation Adds

Treating Doughnut Economics as an ASP constraint-satisfaction problem does two things a purely descriptive or optimization account would not:

- ① It forces every implicit ethical commitment to become **explicit and verifiable**: treating floor and ceiling as hard constraints, not terms in a weighted objective function, is a direct, auditable encoding of sufficientarianism and strong sustainability.
- ② When the problem is infeasible, the model does not stop at a failure flag: it produces a **quantified account** of which constraints remain out of reach, and by how much, turning a solver result into something a policy discussion can actually use.

The Robust Finding: Orders of Magnitude

- The most robust finding is the **order of magnitude** across the three cases, more stable than the single feasibility verdict, which depends on assumed lever magnitudes: the US residual violation (**2815**) is about **two orders of magnitude** above Costa Rica's pre-L14 figure (**77**).
- **Vietnam** shows the problem is comfortably solvable under non-extreme conditions.
- **Costa Rica** shows an intermediate case, solvable only with a more ambitious lever, and even then via several institutionally distinct paths whose final choice remains political.
- **United States** shows the structural limit of marginal reform alone when ecological overshoot is on a much larger scale than the social deficit.

Model Limitations (1/2)

It is worth presenting the limits directly, as part of the argument: they are conscious modeling choices, each motivated, rather than blind spots discovered later.

- **Lever magnitudes are assumptions, not estimates.** The representativeness check bounds them to real cross-country variation, but does not make them causal. A different, equally defensible, set of magnitudes could make Costa Rica infeasible again, or the US implausibly solvable. What survives this uncertainty is the **order of magnitude** between cases, far more robust than any single verdict.
- **The model is static.** It compares a baseline year to a post-reform state, saying nothing about transition dynamics, lever implementation timing, or path-dependencies between levers (e.g. renewables becoming cheaper after circular economy has already cut material demand). A dynamic extension would need a temporal ASP formulation or a hybrid with system-dynamics simulation.
- **Lever effects are additive.** The code sums each selected lever's effect with no interaction terms, while combined policies can have synergies or redundancies in reality. This simplification keeps the best-effort diagnostic model well-defined and fast to compute, at the cost of realism.

Model Limitations (2/2)

- A deliberate choice: it reports the explanation in units an ethical argument can use, such as residual floor or ceiling violation, rather than a set of conflicting logical rules that would be harder to interpret substantively. A minimal-core extraction via `clingo`'s assumption-based mechanisms was considered and set aside for the added complexity of handling conflicts/2 inside an assumption set; it remains a natural next step for future work.
- Both floor and ceiling are national averages: a country that meets an income threshold on average can still have a large share of its population below it. The model inherits this limit directly from the underlying O'Neill et al. dataset.
- **Levers are binary.** Each lever is either fully active or inactive, applied at one single pre-fixed intensity rather than at a graduated level. The carbon-fee lever 114, for example, encodes only one fee rate, whereas a real carbon fee could be set anywhere between roughly €20/t and €150/t. This binary design trades realism for a search space small enough to be exhaustively enumerated.
- **The policy family is limited.** The 14 levers represent one particular policy package, not the entire space of possible options: alternatives such as nuclear power, carbon capture, or efficiency/trade-driven growth are absent. An infeasible result therefore means "infeasible with this menu," not "infeasible in principle."